1 (a) Use standard results from the list of formulae (MF19) to find $\sum_{r=1}^{n}\left(r^3 - r\right)$ in terms of n, fully factorising your answer. [3]

(b) Express $\dfrac{r+3}{r^3-r}$ in the form $\dfrac{A}{r-1}+\dfrac{B}{r}+\dfrac{C}{r+1}$, where A, B and C are constants to be determined, and hence use the method of differences to find $\displaystyle\sum_{r=2}^{n}\frac{r+3}{r^3-r}$. [5]

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(c) Deduce the value of $\displaystyle\sum_{r=2}^{\infty}\frac{r+3}{r^3-r}$. [1]

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